

SAS Portfolio

Brianna Ly

Florida State University

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STA3024: SAS for Data and Statistical Analyses

**Assignment 10: Iterative Processing and Random Number
Generation**

```
LIBNAME Orion '/home/u63559267/my_shared_file_links/jhshows0/STA3024';
```

```
/*Question 1*/
```

```
/*Accumulating totals*/
```

```
PROC CONTENTS DATA= Orion.order_fact;
```

```
RUN;
```

```
PROC PRINT DATA= Orion.order_fact;
```

```
RUN;
```

```
/*Part a*/
```

```
DATA work.order_fact2;
```

```
SET Orion.order_fact;
```

```
Retain acc_total 0;
```

```
acc_total= SUM(acc_total, Total_Retail_Price);
```

```
RUN;
```

```
/*Part b*/
```

```
PROC PRINT DATA= work.order_fact2;
```

```
FORMAT acc_total dollar11.2;
```

```
RUN;
```

Obs	Customer_ID	Employee_ID	Street_ID	Order_Date	Delivery_Date	Order_ID	Order_Type	Product_ID	Quantity	Total_Retail_Price	CostPrice_Per_Unit	Discount	acc_total
1	63	121039	9260125492	11JAN2003	11JAN2003	1230058123	1	220101300017	1	\$16.50	\$7.45	.	\$16.50
2	5	99999999	9260114570	15JAN2003	19JAN2003	1230080101	2	230100500026	1	\$247.50	\$109.55	.	\$264.00
3	45	99999999	9260104847	20JAN2003	22JAN2003	1230106883	2	240600100080	1	\$28.30	\$8.55	.	\$292.30
4	41	120174	1600101527	28JAN2003	28JAN2003	1230147441	1	240600100010	2	\$32.00	\$6.50	.	\$324.30
5	183	120134	1600100760	27FEB2003	27FEB2003	1230315085	1	240200200039	3	\$63.60	\$8.80	.	\$387.90
6	79	99999999	9260101874	02MAR2003	03MAR2003	1230333319	2	240100400005	1	\$234.60	\$115.95	.	\$622.50
7	23	99999999	9260126679	03MAR2003	08MAR2003	1230338566	2	240800200062	1	\$35.40	\$16.15	.	\$657.90
8	23	99999999	9260126679	03MAR2003	08MAR2003	1230338566	2	240800200063	2	\$73.80	\$16.85	.	\$731.70
9	45	99999999	9260104847	09MAR2003	11MAR2003	1230371142	2	240500100004	2	\$127.00	\$28.00	.	\$858.70
10	45	99999999	9260104847	09MAR2003	11MAR2003	1230371142	2	240500200003	1	\$23.20	\$11.10	.	\$881.90
11	56	121059	9260111871	15MAR2003	15MAR2003	1230404278	1	220200300002	2	\$75.00	\$17.05	.	\$956.90
12	183	120149	1600100760	22MAR2003	22MAR2003	1230440481	1	230100600005	1	\$129.80	\$63.20	.	\$1,086.70
13	16	99999999	3940105865	24MAR2003	26MAR2003	1230450371	2	230100600018	2	\$128.40	\$32.20	.	\$1,215.10
14	79	99999999	9260101874	24MAR2003	25MAR2003	1230453723	2	240500200056	1	\$24.10	\$11.55	.	\$1,239.20
15	183	120134	1600100760	25MAR2003	25MAR2003	1230455630	1	240200100233	2	\$91.80	\$22.45	.	\$1,331.00
16	2788	99999999	9050100001	28MAR2003	30MAR2003	1230478006	2	230100100025	2	\$60.60	\$13.15	.	\$1,391.60
17	20	121066	9260118934	01APR2003	01APR2003	1230498538	1	230100300006	1	\$68.50	\$34.35	.	\$1,460.10
18	70046	99999999	2600100017	02APR2003	03APR2003	1230500669	3	240200100131	2	\$148.60	\$41.35	.	\$1,608.70
19	12386	99999999	4750100001	02APR2003	03APR2003	1230503155	2	220101400310	1	\$31.80	\$14.20	.	\$1,640.50
20	23	99999999	9260126679	18APR2003	23APR2003	1230591673	2	220200200024	3	\$178.50	\$32.25	.	\$1,819.00

First used a DATA step to reorder the data based on order_fact, and also created a new variable called acc_total, which is the total sum of the variable total_retail_price. In part b, I adjusted how the output is formatted, adjusting acc_total to have a dollar sign and only have 2 decimal places.

The Payroll department wants to project total employee costs (wages, retirement benefits, and medical benefits) through future years based on assumed increases:

- wages will increase by 3% each year
- retirement benefit costs will increase by 1.4% each year
- medical benefit costs will increase by 9.5% each year.

The present total costs for these items are:

- Wages total \$12,874,000
- Retirement benefit costs total is \$1,765,000
- Medical benefit costs total is \$649,000

/*Question 2*/

/*DO Loops*/

/*Part a*/

DATA work.future_costs (**DROP**=i);

Wages= **12874000**;

Retirement= **1765000**;

Medical= **649000**;

Rate_wage= **0.03**;

Rate_retire= **0.014**;

Rate_med= **0.095**;

DO i=**1** **TO** **10**;

Wages + (Wages*Rate_wage);

Retirement + (Retirement*Rate_retire);

Medical + (Medical*Rate_med);

TotalCost + (Wages+Retirement+Medical);

OUTPUT;

END;

RUN;

PROC PRINT DATA= work.future_costs;

RUN;

Obs	Wages	Retirement	Medical	Rate_wage	Rate_retire	Rate_med	TotalCost
1	13260220.00	1789710.00	710655.00	0.03	0.014	0.095	15760585.00
2	13658026.60	1814765.94	778167.23	0.03	0.014	0.095	32011544.77
3	14067767.40	1840172.66	852093.11	0.03	0.014	0.095	48771577.94
4	14489800.42	1865935.08	933041.96	0.03	0.014	0.095	66060355.39
5	14924494.43	1892058.17	1021680.94	0.03	0.014	0.095	83898588.94
6	15372229.27	1918546.99	1118740.63	0.03	0.014	0.095	102308105.83
7	15833396.14	1945406.64	1225020.99	0.03	0.014	0.095	121311929.61
8	16308398.03	1972642.34	1341397.99	0.03	0.014	0.095	140934367.96
9	16797649.97	2000259.33	1468830.80	0.03	0.014	0.095	161201108.05
10	17301579.47	2028262.96	1608369.72	0.03	0.014	0.095	182139320.20

Performed a DO Loop to calculate total costs for the next 10 years. We included totals for variables wages, retirement, medicine as well as their rates. Total cost is defined by totaling the sum of wages, retirement, and medical expenses.

/*Part b*/

DATA work.future_costs (**DROP**=i);

Wages= **12874000**;

Retirement= **1765000**;

Medical= **649000**;

Rate_wage= **0.03**;

Rate_retire= **0.014**;

Rate_med= **0.095**;

Income= **50000000**;

Rate_income= **0.01**;

DO i=**1** **TO** **10** **UNTIL** (TotalCost > Income);

Wages + (Wages*Rate_wage);

Retirement + (Retirement*Rate_retire);

Medical + (Medical*Rate_med);

TotalCost + (Wages+Retirement+Medical);

Income + (Income*Rate_income);

OUTPUT;

END;

RUN;

PROC PRINT DATA= work.future_costs;

RUN;

Obs	Wages	Retirement	Medical	Rate_wage	Rate_retire	Rate_med	Income	Rate_income	TotalCost
1	13260220.00	1789710.00	710655.00	0.03	0.014	0.095	50500000.0	0.01	15760585.00
2	13658026.60	1814765.94	778167.23	0.03	0.014	0.095	51005000.0	0.01	32011544.77
3	14067767.40	1840172.66	852093.11	0.03	0.014	0.095	51515050.0	0.01	48771577.94
4	14489800.42	1865935.08	933041.96	0.03	0.014	0.095	52030200.5	0.01	66060355.39

Wrote a new dataset that modifies the one in part a to instead examine how many years it will be until the total costs exceed the The present corporate income of \$50,000,000.

A company's income last year was \$50,000,000 and expenses totaled \$38,750,000. Income is projected to increase at one percent per year and expenses are expected to increase at two percent per year

/*Question 3*/

/*DO statement with Conditional Clause*/

/*Part a*/

```
DATA work.expenses (DROP=i);
    Income= 50000000;
    Expense= 38750000;
    Rate_income= 0.01;
    Rate_expense= 0.02;
    DO i=1 TO 100 UNTIL (Expense>Income OR i>30);
        Income + (Income*Rate_income);
        Expense + (Expense*Rate_expense);
    OUTPUT;
    END;
RUN;
```

/*Part b*/

```
PROC PRINT DATA= work.expenses;
    FORMAT Income Dollar15.2 Expense Dollar15.2;
RUN;
```

Obs	Income	Expense	Rate_income	Rate_expense
1	\$50,500,000.00	\$39,525,000.00	0.01	0.02
2	\$51,005,000.00	\$40,315,500.00	0.01	0.02
3	\$51,515,050.00	\$41,121,810.00	0.01	0.02
4	\$52,030,200.50	\$41,944,246.20	0.01	0.02
5	\$52,550,502.51	\$42,783,131.12	0.01	0.02
6	\$53,076,007.53	\$43,638,793.75	0.01	0.02
7	\$53,606,767.61	\$44,511,569.62	0.01	0.02
8	\$54,142,835.28	\$45,401,801.01	0.01	0.02
9	\$54,684,263.63	\$46,309,837.03	0.01	0.02
10	\$55,231,106.27	\$47,236,033.77	0.01	0.02
11	\$55,783,417.33	\$48,180,754.45	0.01	0.02
12	\$56,341,251.51	\$49,144,369.54	0.01	0.02
13	\$56,904,664.02	\$50,127,256.93	0.01	0.02
14	\$57,473,710.66	\$51,129,802.07	0.01	0.02
15	\$58,048,447.77	\$52,152,398.11	0.01	0.02
16	\$58,628,932.25	\$53,195,446.07	0.01	0.02
17	\$59,215,221.57	\$54,259,354.99	0.01	0.02
18	\$59,807,373.78	\$55,344,542.09	0.01	0.02
19	\$60,405,447.52	\$56,451,432.94	0.01	0.02
20	\$61,009,502.00	\$57,580,461.59	0.01	0.02
21	\$61,619,597.02	\$58,732,070.83	0.01	0.02
22	\$62,235,792.99	\$59,906,712.24	0.01	0.02
23	\$62,858,150.92	\$61,104,846.49	0.01	0.02
24	\$63,486,732.43	\$62,326,943.42	0.01	0.02
25	\$64,121,599.75	\$63,573,482.29	0.01	0.02
26	\$64,762,815.75	\$64,844,951.93	0.01	0.02

Used an iterative DO statement with a conditional clause. A command was written to stop the loop once either expenses exceeded income, or once 30 years have been calculated. Data was formatted to include the dollar sign and 2 decimal places.

/*Question 4*/

/*Arrays for Repetitive Computations*/

```
PROC CONTENTS DATA= Orion.orders_midyyear;  
RUN;
```

```
PROC PRINT DATA= Orion.orders_midyyear;  
RUN;
```

/*Part a*/

```
DATA discount_sales;  
SET Orion.orders_midyyear;  
KEEP Customer_ID Month1-Month6;  
ARRAY MON{6} Month1-Month6;  
    DO i=1 TO 6;  
        Mon{i}=SUM(Mon{i},-(Mon{i}*0.05));  
    END;  
    OUTPUT;  
RUN;
```

/*Part b*/

```
PROC PRINT DATA= discount_sales;  
    FORMAT Month1- Month6 Dollar10.2;  
RUN;
```


Obs	Customer_ID	Month1	Month2	Month3	Month4	Month5	Month6
1	5	\$202.45	.	\$454.10	\$499.51	\$374.63	\$182.20
2	10	\$178.69	\$393.39	\$2,733.06	\$3,006.36	\$2,254.77	\$160.83
3	11	\$74.29	.	.	.	.	\$66.86
4	12	\$128.82	.	\$111.72	\$122.89	\$92.17	\$115.94
5	18	.	.	\$27.93	\$30.72	\$23.04	.
6	24	\$88.35	\$252.51	.	.	.	\$79.52
7	27	\$295.17	\$743.76	.	.	.	\$265.65
8	31	\$1,410.09	\$278.64	.	.	.	\$1,269.08
9	34	\$610.38	.	\$81.99	\$90.18	\$67.64	\$549.34
10	41	\$127.30	\$113.24	\$297.35	\$327.09	\$245.31	\$114.57
11	45	\$421.69	\$205.39	\$38.19	\$42.01	\$31.51	\$379.52
12	49	\$23.56	.	.	.	.	\$21.20
13	53	.	\$380.38	\$171.38	\$188.52	\$141.39	.
14	56	.	\$36.01	.	.	.	.
15	65	.	.	\$333.83	\$367.21	\$275.41	.
16	69	\$22.33	\$3.04	.	.	.	\$20.09
17	89	\$199.31	\$500.46	.	.	.	\$179.38
18	90	\$31.92	\$104.69	\$377.06	\$414.76	\$311.07	\$28.73
19	92	\$16.06	.	\$152.48	\$167.72	\$125.79	\$14.45
20	111	.	\$662.72	.	.	.	.
21	171	\$70.29	\$507.87	\$1,179.33	\$1,297.26	\$972.95	\$63.26
22	183	.	\$139.75	.	.	.	.
23	215	.	.	\$57.76	\$63.54	\$47.65	.
24	908	.	.	\$493.34	\$542.67	\$407.00	.

Created a new data set titled discount_sales to see how a 5% price discount would affect income by months. I used an array to identify the variables as months 1 to 6 to be processed as a group, and used a DO loop to apply the 5% discount that would adjust the monthly data. Data was formatted to include the dollar sign and 2 decimal places.

/*Question 5*/

/*Generating Random Values*/

/*Part a*/

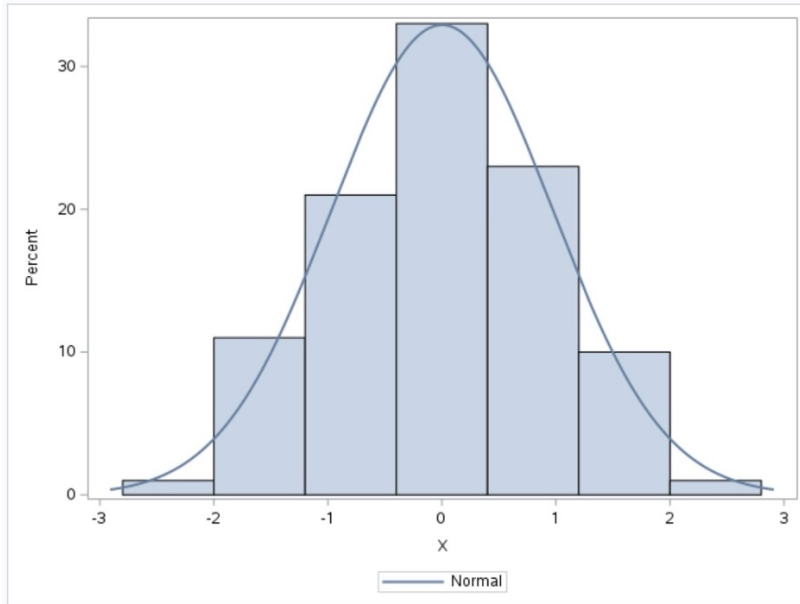
```
DATA Five_a;  
    DO I=1 to 100;  
        X= RAND('normal');  
    OUTPUT;  
    END;  
RUN;
```

```
PROC SGPLOT DATA= Five_a;  
HISTOGRAM X;  
DENSITY X;  
RUN;
```

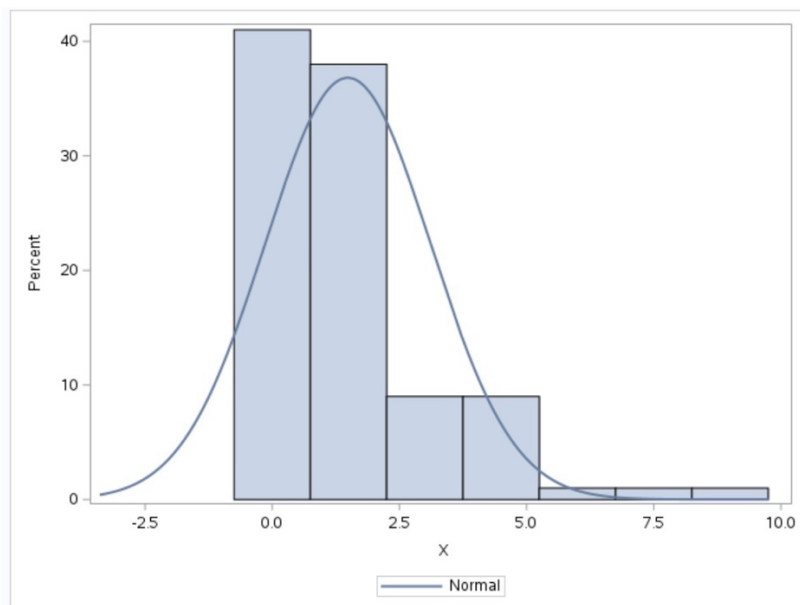
/*Part b*/

```
DATA Five_b;  
    DO I=1 to 100;  
        X= RAND('LOGNormal');  
    OUTPUT;  
    END;  
RUN;
```

```
PROC SGPLOT DATA= Five_b;  
HISTOGRAM X;  
DENSITY X;  
RUN;
```



In part a, I generated 100 random values from a random normal distribution, then created a histogram with these random values with an overlay of a standard normal density curve. The generated data seems to closely follow a standard normal distribution.



In part b, 100 random values were also generated, but following a log normal distribution, then used PROC SGPLOT to graph the histogram and density curve.

STA3064: Introduction to Statistical Modeling with SAS

Project B: ANOVA AND ANCOVA Analysis on Sleep

Apnea

Motivation

Obstructive sleep apnea is a sleeping disorder characterized by pauses in breathing while asleep. This condition includes symptoms such as excessive daytime sleepiness, snoring, high blood pressure, and more (Mayo Clinic, 2023). Fitting a statistical model for this data could offer insight as to how blood oxygen level and sleepiness could affect a person's diagnosis of Obstructive Sleep Apnea (OSA). This project is meant to identify the relationship between the variables, and to what degree the correlation exists. The data from this project can open research questions regarding how blood oxygen levels in blood affect OSA, and even raise questions in neuroscience, such as identifying how these blood oxygen levels affect the brain and how this relates to daytime sleepiness.

Data Description

The dataset was created by Paul M. Macey in 2023 with the intent of exploring the relationship between OSA and stress. In a related publication titled "Stress in Obstructive Sleep Apnea" by Jasmine L. Wong, Macey, and other researchers, sought to describe the relationship between stress in relation to other psychological symptoms. Inside the dataset, there are a total of 58 observations, and for this project, three of the variables will be explored. The response variable is Apnea-Hypopnea Index (AHI), a measure of the severity of OSA in an individual. The predictor variables consist of a categorical variable and a continuous variable. The categorical variable is the Epworth sleepiness scale (ESS), a scale that measures daytime sleepiness. Because the range of values on this scale goes from 0 to 24, in this project, this will be modified and collapsed to better include all the observations into categories using a general interpretation of ESS scores. The continuous variable is Average oxygen saturation of arterial blood (SaO2avg) which helps determine how much oxygen is in a person's blood.

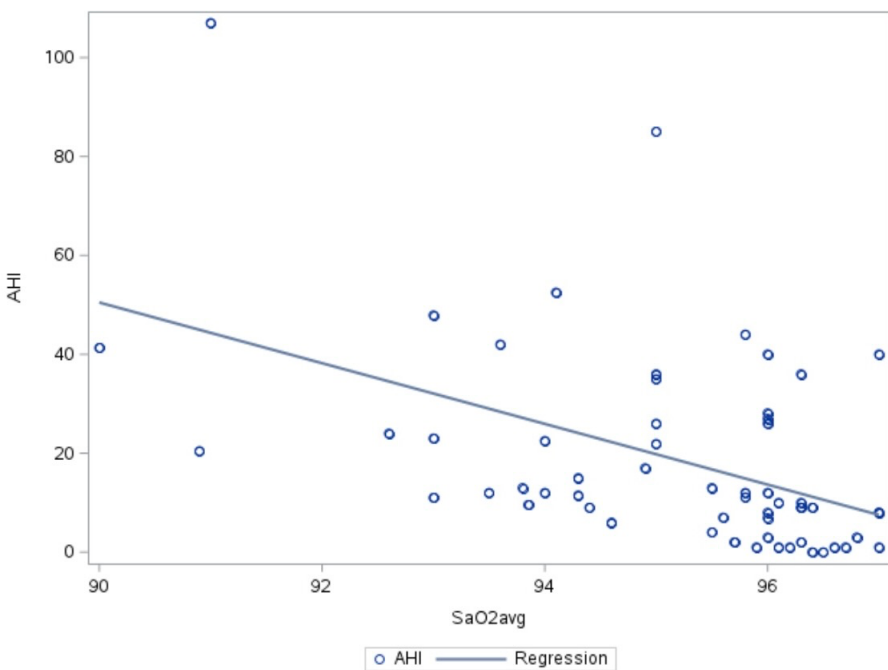
Data Exploration

Code to import dataset

```
PROC IMPORT DATAFILE= '/home/u63559267/BQL23_3064/Datasets/Raw data -  
OSA-CON with sleep study.xlsx'  
    OUT= Sleep  
    DBMS= XLSX  
    REPLACE;  
    GETNAMES= yes;  
    RANGE="Sheet1$A2:0";  
RUN;  
PROC PRINT DATA= work.Sleep;  
RUN;
```

The dataset is then further adjusted to reflect the variables used in this project. For the variable ESS, the numerical values have been collapsed and renamed to five general ESS score interpretations. The new dataset will only show the necessary variables.

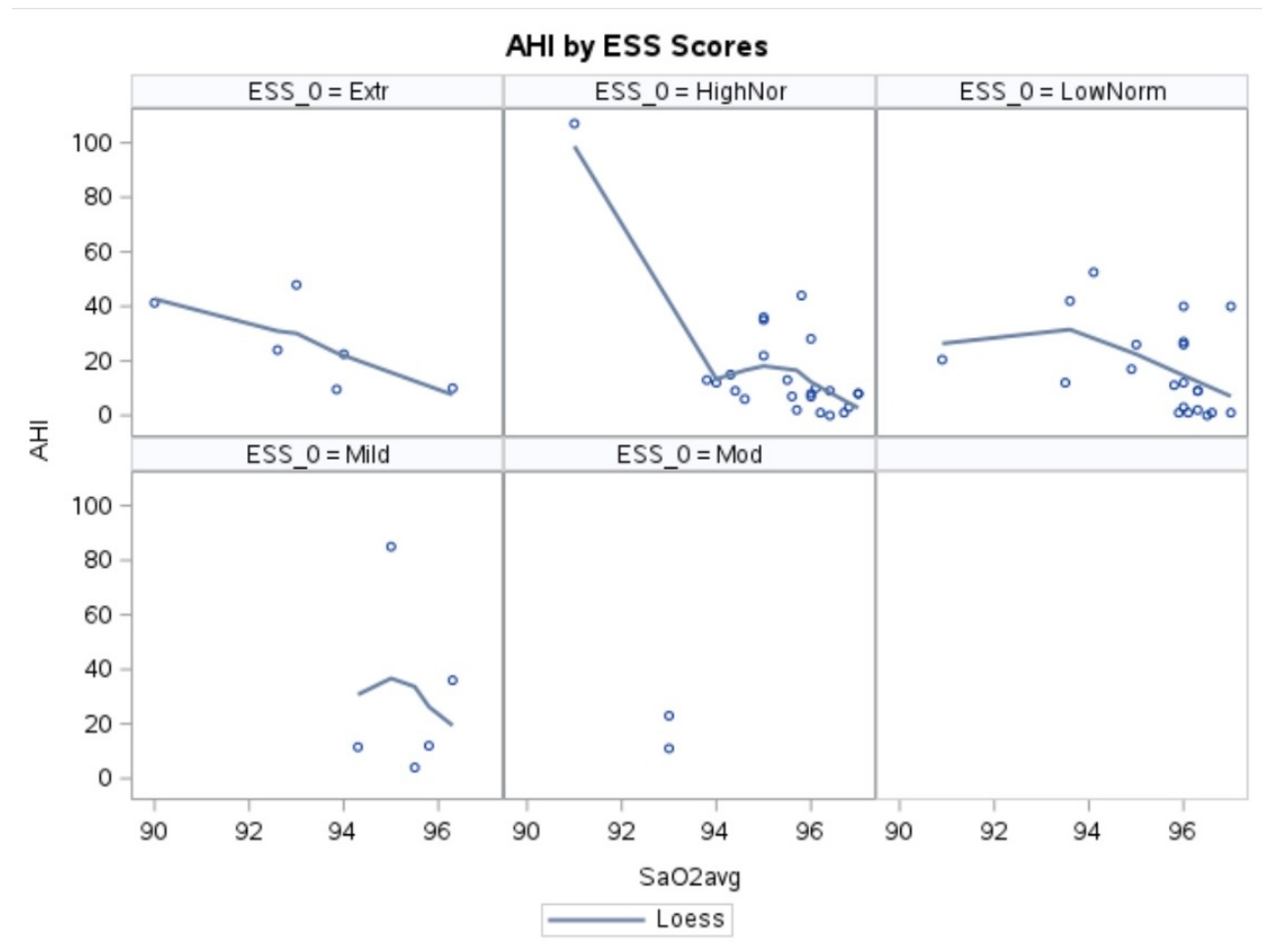
Scatterplot for relationship between variables with all ESS



This scatterplot shows an overall negative correlation between SaO2avg and AHI, most of the datapoints clustered near bottom right corner of the graph which shows a high SaO2avg and a

low AHI. These points appear to be close to the regression line, but as AHI increased, the datapoints grow further from the regression line, suggesting that there could be outliers.

Scatterplots for relationships between variables by different ESS



From the graphs, one noticeable flaw is that the number of observations for anything with an ESS of Moderate or Mild is very small and can easily skew the data if we were to measure them on their own. Many of the datapoints fall in either HighNor (high normal) or LowNorm (low normal) ESS scores. We can also see a steep descent for the LOESS graph for ESS score of High normal, but only one datapoint which has caused this. It suggests that the one point could be an outlier in the data.

Model Fitting and Analysis

ANOVA Modeling

The GLM Procedure

Dependent Variable: AHI AHI

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	1076.76835	269.19209	0.63	0.6420
Error	53	22581.00786	426.05675		
Corrected Total	57	23657.77621			

R-Square	Coeff Var	Root MSE	AHI Mean
0.045514	109.3620	20.64114	18.87414

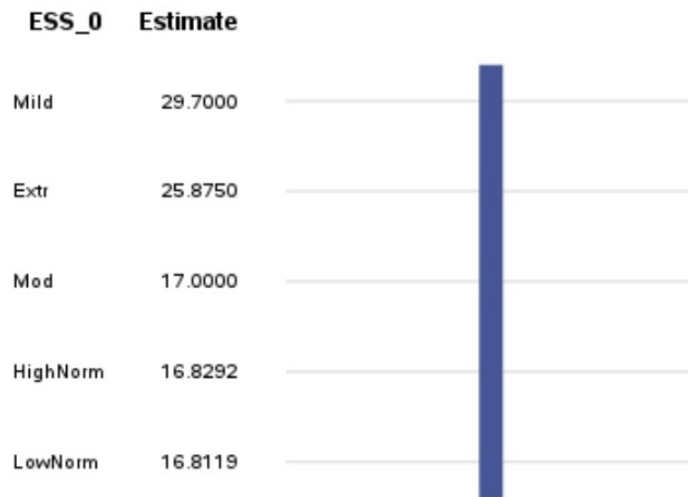
Source	DF	Type III SS	Mean Square	F Value	Pr > F
ESS_0	4	1076.768350	269.192087	0.63	0.6420

One-way ANOVA model with only categorical variable ESS.

The P value is 0.6420, which is less than a standard significance value of 0.05, which means that we should not reject the null hypothesis, which would suggest that the data is not statistically significant, and we cannot reject the idea that the means in the separate categorizes are different.

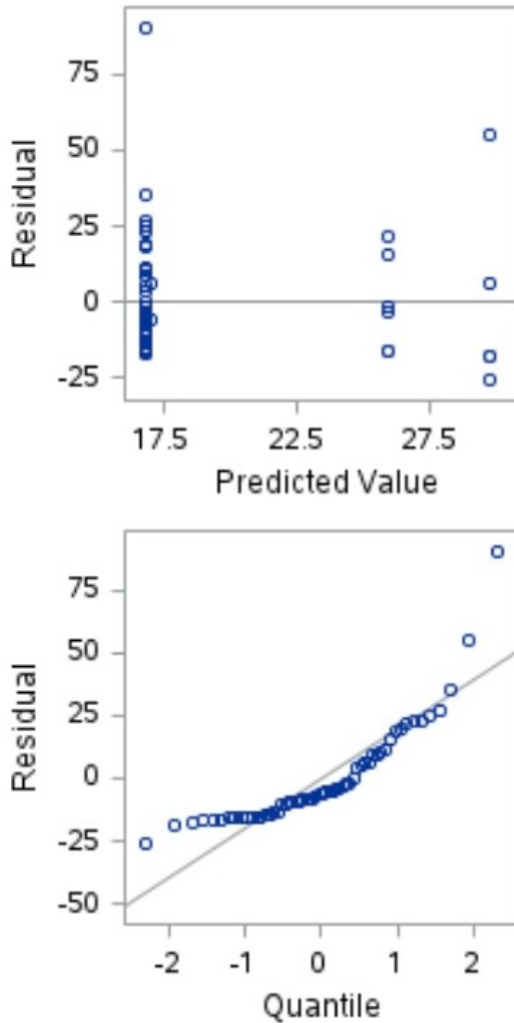
AHI Tukey Grouping for Means of ESS_0 (Alpha = 0.05)

Means covered by the same bar are not significantly different.



Tukey Grouping means to identify differences.

Using the Tukey Grouping Means, we are able to see that the means for each category of ESS is not individually significant, suggesting that the means of AHI remain relatively the same no matter the ESS.



Residual plots

In the predicted residual plot, there doesn't appear to be any noticeable pattern by the datapoints, which could suggest linearity in the data. The variation appears to be consistent, so we might be able to assume constant variance. The Q-Q plot appears to be mostly linear, closely following along the line, which suggests that it follow a normal pattern. Overall, we can conclude that the conditions have been met.

ANCOVA Modeling

The GLM Procedure

Dependent Variable: AHI AHI

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	8	8355.75639	1044.46955	3.34	0.0039
Error	49	15302.01981	312.28612		
Corrected Total	57	23657.77621			

R-Square	Coeff Var	Root MSE	AHI Mean
0.353193	93.62875	17.67162	18.87414

Source	DF	Type III SS	Mean Square	F Value	Pr > F
SaO2avg	1	1111.635607	1111.635607	3.56	0.0651
ESS_0	3	1579.666288	526.555429	1.69	0.1822
SaO2avg*ESS_0	3	1578.816243	526.272081	1.69	0.1824

ANCOVA Model

The p-value of SaO2avg is 0.0651, which is slightly more than a standard significance value of 0.05 but suggests that it is not statistically significant. The variable ESS_0 and the interaction are approximately around the same with p values of 0.1822 and 0.1824, and both are not statistically significant. The overall model, however, has a p-value of 0.003, which implies that it is statistically significant. In this case, it might imply that there is multicollinearity within the model.

The GLM Procedure

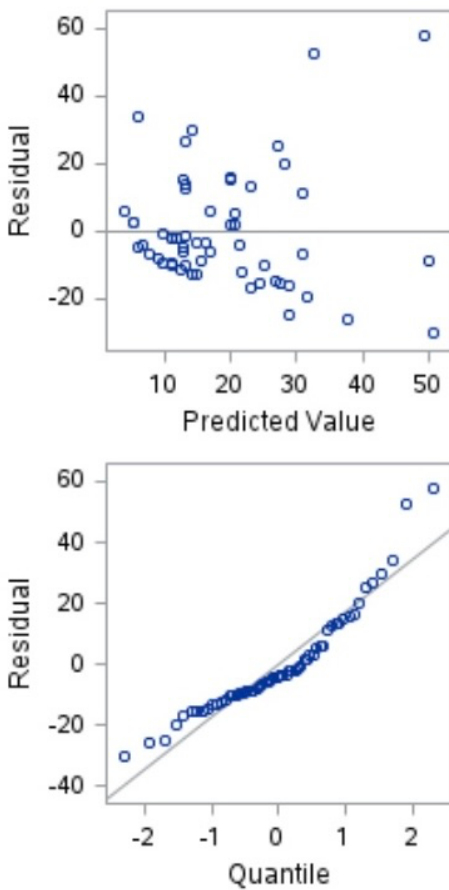
Dependent Variable: AHI AHI

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	6776.94015	1355.38803	4.18	0.0029
Error	52	16880.83606	324.63146		
Corrected Total	57	23657.77621			

R-Square	Coeff Var	Root MSE	AHI Mean
0.286457	95.46148	18.01753	18.87414

Source	DF	Type III SS	Mean Square	F Value	Pr > F
SaO2avg	1	5700.171801	5700.171801	17.56	0.0001
ESS_0	4	1474.884472	368.721118	1.14	0.3500

Looking at an ANCOVA model without the interaction, it seems to be a better model. The SaO2avg has a p-value of 0.0001, which implies that they are statistically significant in the model. The ESS still remains to be not significant, but the p-value of the overall model 0.0029, which is less than 0.05, and statistically significant.



Residual Plots

In the top plot, there is not any clear patten that stands out, which implies that the data holds linearity. The variation also appears to remain consistent, which means that there is constant variance. The Q-Q plot appears relatively linear, which means that it likely follows a normal pattern. We can conclude, overall, that the conditions have been met.

Comparison

The GLM Procedure					
Dependent Variable: AHI AHI					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	6776.94015	1355.38803	4.18	0.0029
Error	52	16880.83606	324.63146		
Corrected Total	57	23657.77621			

R-Square	Coeff Var	Root MSE	AHI Mean
0.286457	95.46148	18.01753	18.87414

Source	DF	Type III SS	Mean Square	F Value	Pr > F
SaO2avg	1	5700.171801	5700.171801	17.56	0.0001
ESS_0	4	1474.884472	368.721118	1.14	0.3500

ANCOVA

The categorical variable alone, did not appear to have any significant effect on the AHI value in the observations and it was not statistically significant. After introducing the quantitative variable SaO2avg in the ANCOVA model, the fit appears to be slightly better than it was in the ANOVA model with just the categorical variable alone.

The overall model in the ANCOVA modeling is also statistically significant, while the ANOVA model does not reflect this. This helps clarify the function of the categorical variable and how it affects the overall model by interacting with the quantitative variable.

Conclusion

The data shows suggests that the ESS does not directly affect the AHI score, which measures for severe the OSA is within an individual. However, when ESS is paired with SaO2avg, the AHI score becomes more dependent on the two variables combined rather than with the ESS score alone. Average oxygen saturation of arterial blood appears to be directly correlated to the severity of OSA. The data suggests that excessive daytime sleepiness has an effect on the severity of OSA, but only when it was paired with the average oxygen saturation of arterial blood levels. This suggests that excessive daytime sleepiness has a different variable cause, or there are other variables that affect the severity of OSA that this project did not

The GLM Procedure					
Dependent Variable: AHI AHI					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	1076.76835	269.19209	0.63	0.6420
Error	53	22581.00786	426.05675		
Corrected Total	57	23657.77621			

R-Square	Coeff Var	Root MSE	AHI Mean
0.045514	109.3620	20.64114	18.87414

Source	DF	Type III SS	Mean Square	F Value	Pr > F
ESS_0	4	1076.768350	269.192087	0.63	0.6420

ANOVA

consider, and if paired with ESS, can affect the response variable respectively. This opens possible research for other factors that affect OSA. A low average oxygen saturation of arterial blood appears to be one of the major factors that contributes to a high OSA severity. This might also be interpreted in which not having enough oxygen traveling to the brain, which might temporarily stop bodily functions that occur during sleep. To further use this data in research, some topics that can be explored might include brain function with oxygen levels, and other variables regarding sleep and brain activity as it relates to OSA.

References

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(2021). Stress in obstructive sleep apnea. *Scientific Reports*, 11(1).

<https://doi.org/10.1038/s41598-021-91996-5>

STA4203: Applied Regression Methods

Assignment 3: Variable and Model Selection

Data is read from a txt document with the following variables:

- Obs: Observation number
- Lcavol: log(cancer volume)
- Lweight: log(prostate weight)
- Age: age
- Lbph: log(benign prostatic hyperplasia amount)
- Svi: seminal vesicle invasion
- Lcp: log(capsular penetration)
- Gleason: Gleason score
- pgg45: percentage Gleason scores 4 or 5
- lpsa: log(prostate specific antigen)

Where Lcavol is our response variable.

```
/*Read in dataset and clean data*/
```

```
DATA prostate_rawdata;
```

```
INFILE '/home/u63559267/Datasets/prostate.txt';
```

```
INPUT Obs Y X1 X2 X3 X4 X5 X6 X7 X8;
```

```
RUN;
```

```
PROC PRINT DATA= prostate_rawdata;
```

```
RUN;
```

```
/*Remove Obs column*/
```

```
DATA prostate_data;
```

```
SET prostate_rawdata (DROP= Obs);
```

```
RUN;
```

```
PROC PRINT DATA= prostate_data;
```

```
RUN;
```

/*Question 1*/

/*Finding best subset model using Forward Selection*/

PROC REG DATA= prostate_data;

MODEL Y= X1 X2 X3 X4 X5 X6 X7 X8/

SELECTION= Forward;

RUN;

Forward Selection: Step 6

Variable X6 Entered: R-Square = 0.6749 and C(p) = 5.5337

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	90.00479	15.00080	31.14	<.0001
Error	90	43.35425	0.48171		
Corrected Total	96	133.35903			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	-2.53592	1.11161	2.50702	5.20	0.0249
X2	0.02124	0.01060	1.93550	4.02	0.0480
X3	-0.08850	0.05319	1.33383	2.77	0.0996
X5	0.34485	0.07238	10.93518	22.70	<.0001
X6	0.21176	0.15023	0.95708	1.99	0.1621
X7	-0.00735	0.00427	1.42978	2.97	0.0884
X8	0.54442	0.07572	24.90005	51.69	<.0001

Bounds on condition number: 2.8876, 67.346

No other variable met the 0.5000 significance level for entry into the model.

Summary of Forward Selection							
Step	Variable Entered	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	X8	1	0.5394	0.5394	32.4276	111.27	<.0001
2	X5	2	0.1061	0.6455	5.5466	28.12	<.0001
3	X2	3	0.0078	0.6533	5.4222	2.09	0.1514
4	X3	4	0.0102	0.6635	4.6462	2.79	0.0984
5	X7	5	0.0043	0.6677	5.4881	1.16	0.2834
6	X6	6	0.0072	0.6749	5.5337	1.99	0.1621

Best Model: Y= X2+ X3+ X5+ X6+ X7+ X8

/*Finding best subset model using Backward Elimination*/

```
PROC REG DATA= prostate_data;
  MODEL Y= X1 X2 X3 X4 X5 X6 X7 X8 /
    SELECTION= Backward;
RUN;
```

Backward Elimination: Step 4

Variable X7 Removed: R-Square = 0.6635 and C(p) = 4.6462

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	88.48063	22.12016	45.35	<.0001
Error	92	44.87840	0.48781		
Corrected Total	96	133.35903			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	-1.21022	0.68213	1.53548	3.15	0.0793
X2	0.02003	0.01033	1.83388	3.76	0.0556
X3	-0.08935	0.05352	1.35942	2.79	0.0984
X5	0.30907	0.06175	12.22216	25.06	<.0001
X8	0.54273	0.07580	25.00533	51.26	<.0001

Bounds on condition number: 1.5068, 21.3

All variables left in the model are significant at the 0.1000 level.

Summary of Backward Elimination							
Step	Variable Removed	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	X1	7	0.0006	0.6762	7.1759	0.18	0.6759
2	X4	6	0.0013	0.6749	5.5337	0.36	0.5494
3	X6	5	0.0072	0.6677	5.4881	1.99	0.1621
4	X7	4	0.0043	0.6635	4.6462	1.16	0.2834

Best Model: $Y = X_2 + X_3 + X_5 + X_8$

/*Finding best subset model using Stepwise Regression*/

PROC REG DATA= prostate_data;

MODEL Y= X1 X2 X3 X4 X5 X6 X7 X8 /

SELECTION= Stepwise;

RUN;

Stepwise Selection: Step 2

Variable X5 Entered: R-Square = 0.6455 and C(p) = 5.5466

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	86.08094	43.04047	85.57	<.0001
Error	94	47.27809	0.50296		
Corrected Total	96	133.35903			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	0.09135	0.20527	0.09960	0.20	0.6573
X5	0.32838	0.06193	14.14282	28.12	<.0001
X8	0.53162	0.07501	25.26330	50.23	<.0001

Bounds on condition number: 1.431, 5.7241

All variables left in the model are significant at the 0.1500 level.

No other variable met the 0.1500 significance level for entry into the model.

Summary of Stepwise Selection								
Step	Variable Entered	Variable Removed	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	X8		1	0.5394	0.5394	32.4276	111.27	<.0001
2	X5		2	0.1061	0.6455	5.5466	28.12	<.0001

Best Model: $Y = X5 + X8$

/*Finding the best subset of the three based on Mallow's CP, adjusted R², AIC, and BIC*/

```
PROC REG DATA=prostate_data OUTEST= EST;
  MODEL Y= X1 X2 X3 X4 X5 X6 X7 X8 / SELECTION= adjrsq SSE AIC BIC
  ADJRSQ CP;
  OUTPUT OUT= out p=p r=r;
RUN;
QUIT;
```

Number in Model	Adjusted R-Square	R-Square	C(p)	AIC	BIC	SSE	Variables in Model
6	0.6532	0.6749	5.5337	-64.1147	-60.7942	43.35425	X2 X3 X5 X6 X7 X8
7	0.6508	0.6762	7.1759	-62.5075	-58.9264	43.17904	X2 X3 X4 X5 X6 X7 X8
7	0.6500	0.6755	7.3658	-62.2988	-58.7545	43.27206	X1 X2 X3 X5 X6 X7 X8
5	0.6495	0.6677	5.4881	-63.9966	-61.1434	44.31133	X2 X3 X5 X7 X8
4	0.6488	0.6635	4.6462	-64.7632	-62.1858	44.87840	X2 X3 X5 X8
6	0.6479	0.6700	6.8825	-62.6480	-59.5520	44.01476	X2 X3 X4 X5 X7 X8
8	0.6475	0.6769	9.0000	-60.7012	-56.8812	43.09290	X1 X2 X3 X4 X5 X6 X7 X8
5	0.6475	0.6658	6.0053	-63.4438	-60.6621	44.56458	X2 X3 X4 X5 X8
6	0.6470	0.6690	7.1323	-62.3789	-59.3237	44.13707	X1 X2 X3 X5 X7 X8
5	0.6465	0.6649	6.2575	-63.1754	-60.4282	44.68808	X2 X5 X6 X7 X8
6	0.6460	0.6682	7.3680	-62.1255	-59.1088	44.25252	X1 X2 X5 X6 X7 X8
5	0.6457	0.6642	6.4535	-62.9674	-60.2469	44.78403	X2 X3 X5 X6 X8
5	0.6457	0.6642	6.4610	-62.9593	-60.2399	44.78774	X1 X2 X3 X5 X8
7	0.6454	0.6713	8.5287	-61.0306	-57.7093	43.84150	X1 X2 X3 X4 X5 X7 X8
6	0.6443	0.6665	7.8195	-61.6421	-58.6986	44.47361	X1 X2 X3 X4 X5 X8
6	0.6441	0.6663	7.8705	-61.5877	-58.6524	44.49854	X2 X3 X4 X5 X6 X8
5	0.6436	0.6622	7.0012	-62.3882	-59.7420	45.05225	X1 X2 X5 X7 X8
6	0.6428	0.6651	8.2020	-61.2345	-58.3525	44.66088	X2 X4 X5 X6 X7 X8
4	0.6427	0.6576	6.2586	-63.0714	-60.6734	45.66798	X2 X5 X7 X8
7	0.6425	0.6685	9.2668	-60.2343	-57.0519	44.20292	X1 X2 X4 X5 X6 X7 X8
6	0.6423	0.6646	8.3307	-61.0977	-58.2363	44.72390	X1 X2 X3 X5 X6 X8
3	0.6421	0.6533	5.4222	-63.8685	-61.6516	46.23782	X2 X5 X8
4	0.6417	0.6567	6.5006	-62.8200	-60.4485	45.78650	X1 X2 X5 X8
5	0.6417	0.6604	7.4862	-61.8781	-59.2972	45.28975	X3 X5 X6 X7 X8
4	0.6414	0.6563	6.6018	-62.7151	-60.3546	45.83606	X5 X6 X7 X8
7	0.6406	0.6668	9.7370	-59.7303	-56.6354	44.43317	X1 X2 X3 X4 X5 X6 X8

Then we sort for each criterion.

```

67  /*CP*/
68  PROC SORT DATA= EST OUT= EST_CP;
69      BY _CP_;
70  PROC PRINT DATA= EST_CP;
71  RUN;
72
73  /*Adjusted R^2*/
74  PROC SORT DATA= EST OUT= EST_adjrsq;
75      BY DESCENDING _ADJRSQ_;
76  PROC PRINT DATA= EST_adjrsq;
77  RUN;
78
79  /*AIC*/
80  PROC SORT DATA= EST OUT=EST_AIC;
81      BY _AIC_;
82  PROC PRINT DATA= EST_AIC;
83  RUN;
84
85  /*BIC*/
86  PROC SORT DATA= EST OUT=EST_BIC;
87      BY _BIC_;
88  PROC PRINT DATA= EST_BIC;
89  RUN;
90

```

According to Mallows' CP, AIC and BIC, model $Y = X_2 + X_3 + X_5 + X_8$ is the best (starting with the full model).

According to AdjRsqr, model $Y = X_2 + X_3 + X_5 + X_6 + X_7 + X_8$ is the best (starting with the intercept model).

STA4173: Biostatistics

Assignment 6&7: Single and Two-factor Studies

1. (50 points total) Four strains of cultured *Staphylococcus aureus* – bacteria that cause staph infections – were observed for 24 hours. The following table reports bacteria counts, in millions, for different cases from each of the four strains:

Strain A	Strain B	Strain C	Strain D
9	3	30	44
27	12	47	38
22	7	50	37
30	15	52	49
16	12	26	40

DATA Bacteria;

INPUT Strain \$ Count @@;

DATALINES;

A 9 B 3 C 30 D 44

A 27 B 12 C 47 D 38

A 22 B 7 C 50 D 37

A 30 B 15 C 52 D 49

A 16 B 12 C 26 D 40

;

RUN;

PROC PRINT DATA= Bacteria;

RUN;

Sorting the data by Strain, we are able to visualize the data in SAS.

PROC SORT DATA= Bacteria;

BY Strain;

RUN;

PROC PRINT DATA= Bacteria;

RUN;

Obs	Strain	Count
1	A	9
2	A	27
3	A	22
4	A	30
5	A	16
6	B	3
7	B	12
8	B	7
9	B	15
10	B	12
11	C	30
12	C	47
13	C	50
14	C	52
15	C	26
16	D	44
17	D	38
18	D	37
19	D	49
20	D	40

/*At the 5% significance level, do the data provide sufficient evidence to conclude that a difference exists in mean bacteria counts among the four strains? Give the hypotheses under both models, the test statistic, P-value, and conclusion.*/

```
PROC GLM DATA= Bacteria;
CLASS Strain;
MODEL Count= Strain;
MEANS Strain / LINES TUKEY SCHEFFE;
RUN;
```

The GLM Procedure					
Dependent Variable: Count					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	3683.400000	1227.800000	18.55	<.0001
Error	16	1058.800000	66.175000		
Corrected Total	19	4742.200000			

R-Square	Coeff Var	Root MSE	Count Mean
0.776728	28.74488	8.134802	28.30000

Source	DF	Type I SS	Mean Square	F Value	Pr > F
Strain	3	3683.400000	1227.800000	18.55	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Strain	3	3683.400000	1227.800000	18.55	<.0001

Cell Means:

$H_0 = \mu_1 = \mu_2 = \mu_3 = \mu_4$ (All group means are equal)

H_a = Not all group means are equal

Treatment effects:

$H_0 = \tau_1 = \tau_2 = \tau_3 = \tau_4 = 0$

H_a = Not all are 0

F-value= 18.55

P-value= <0.0001

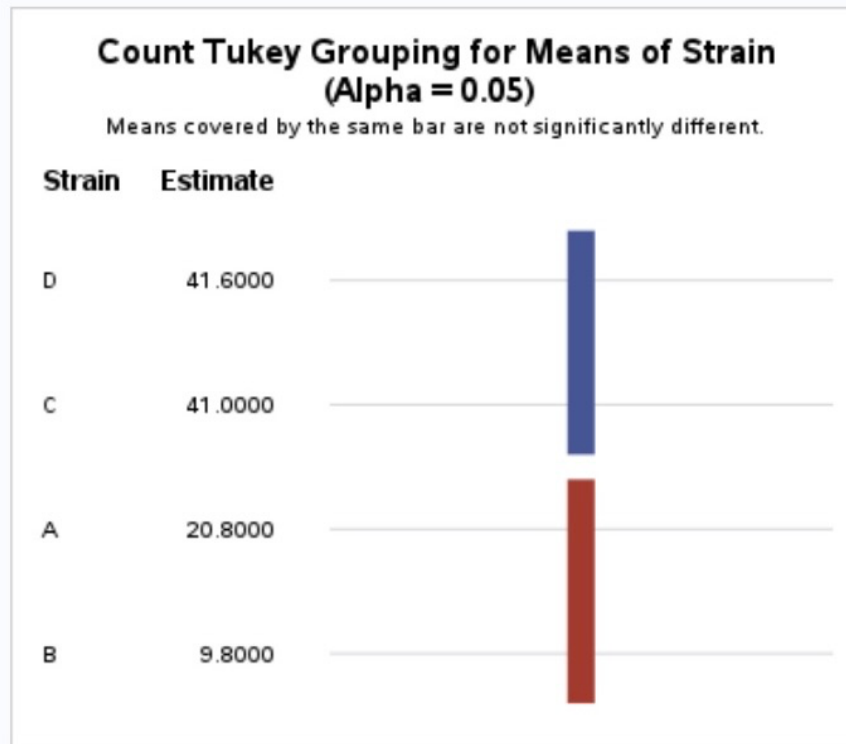
Conclusion: Because the p-value is less than our standard sig value, we have enough evidence to reject the null hypothesis and conclude that some treatment means are significantly different and that the different strains affect bacteria count.

The GLM Procedure

Tukey's Studentized Range (HSD) Test for Count

Note: This test controls the Type I experimentwise error rate, but it generally has a higher Type II error rate than REGWQ.

Alpha	0.05
Error Degrees of Freedom	16
Error Mean Square	66.175
Critical Value of Studentized Range	4.04606
Minimum Significant Difference	14.72



According to Tukey's method, we see that strain A and B have significantly different means from strains C and D. Means C and D appear to be higher than means A and B

```

/*Question 2*/

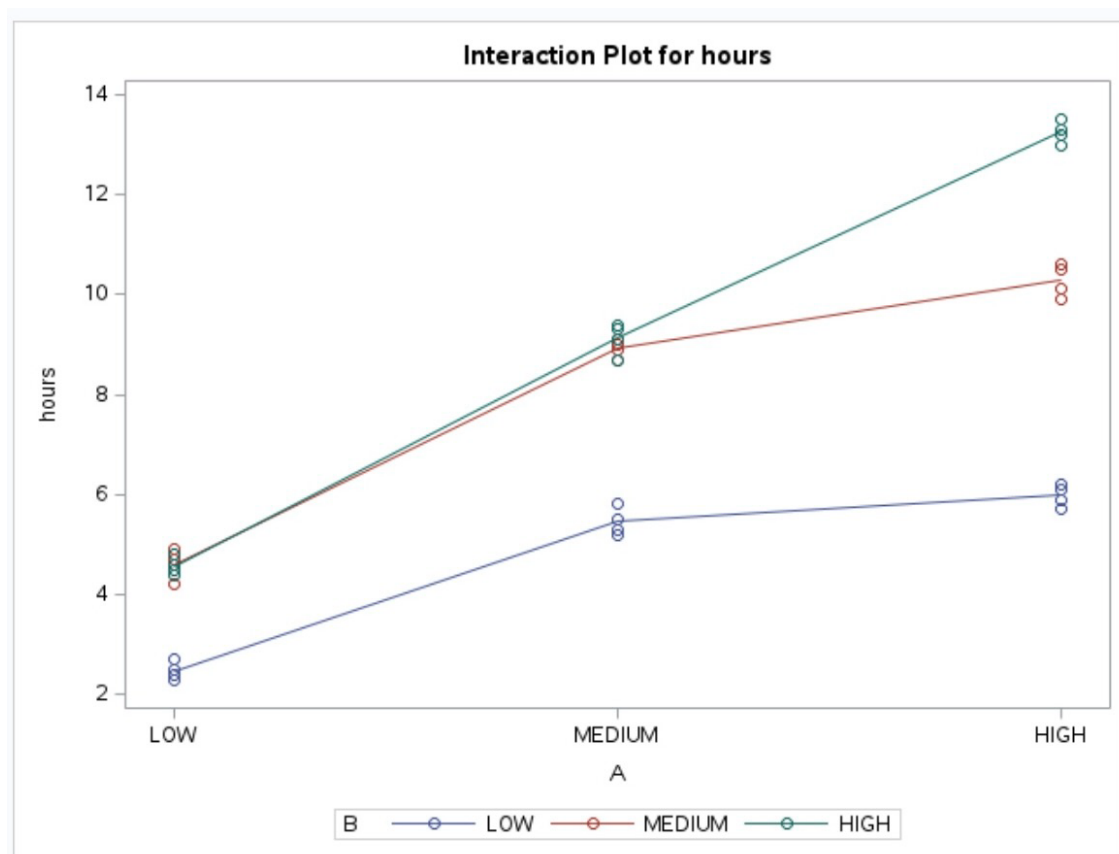
/*Datastep*/
FILENAME Fever '/home/u63559267/my_shared_file_links/jhshows0/Data Sets/fever.txt';
DATA Fever;
INFILE Fever;
INPUT Hours A $ B $ @@;
RUN;

PROC PRINT DATA= Fever;
RUN;

/*Part a*/
/*Create an interaction plot. Does there appear to be significant interaction between the levels of
ingredient A and ingredient B? Why or why not?*/

PROC GLM DATA= Fever ORDER= DATA;
CLASS A B;
MODEL Hours= A|B;
LSMEANS A*B/ LINES ADJUST= Tukey;
RUN;

```



Creating an interaction plot, we are able to observe a significant interaction between the levels of ingredient A and ingredient B. From the plot, in factor B, low line appears to be relatively parallel to the other lines. However, the Medium and High lines appear to intersect, looking like

they are almost layered on each other. From this we can assume there is an interaction. This means that the change in A depends on the level of B.

The GLM Procedure					
Dependent Variable: Hours					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	8	373.1050000	46.6381250	774.91	<.0001
Error	27	1.6250000	0.0601852		
Corrected Total	35	374.7300000			

R-Square	Coeff Var	Root MSE	Hours Mean
0.995664	3.415221	0.245327	7.183333

Source	DF	Type I SS	Mean Square	F Value	Pr > F
A	2	220.0200000	110.0100000	1827.86	<.0001
B	2	123.6600000	61.8300000	1027.33	<.0001
A*B	4	29.4250000	7.3562500	122.23	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
A	2	220.0200000	110.0100000	1827.86	<.0001
B	2	123.6600000	61.8300000	1027.33	<.0001
A*B	4	29.4250000	7.3562500	122.23	<.0001

Using the treatment effects model, a hypothesis test was performed to test for an interaction effect between the levels of ingredient A and B.

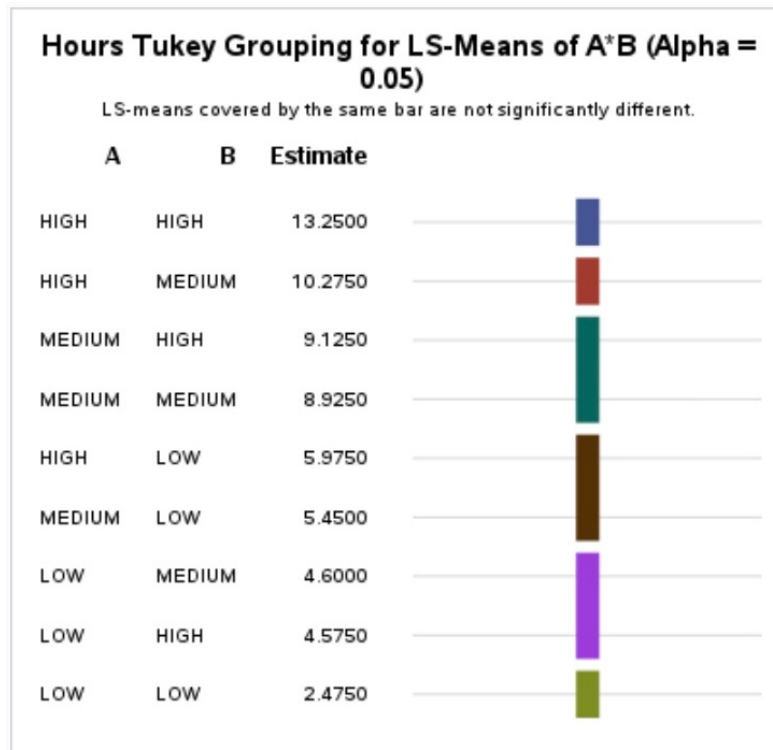
H_0 = All interactions = 0

H_a : Not all= 0

F-value= 122.23

P-value= < 0.0001

Conclusion: Because our p-value is smaller than the significance level, we have enough evidence to reject the null hypothesis and conclude that there is a significant interaction effect.



We compare the change in the mean of factor A as its level increases for each level of B because we already found a significant interaction between the two factors.

Using Tukey's method:

For Factor B- Low, we see there is an increase of about double in estimated hours as Factor A goes from low to medium, then a very slight increase as it goes from medium to high. (2.475 -> 5.45 -> 5.975)

For Factor B- Medium, it follows a similar pattern, with an increase of estimate of nearly double from low to medium, but a larger increase from Med to High. (4.6 -> 8.925 -> 10.275)

For Factor B- High, it follows the same pattern with a slightly higher increase as it goes from medium to high. (4.575 -> 9.125 -> 13.25)

It looks like while the general increasing pattern remains, the effect of factor A is stronger when B is at a Medium or High level, but slightly weaker when B is Low. Overall, it appears as though the impact of factor A on the response (hours) depends on factor B, and vice versa.